Deciding the Trig Rule

Exam code: 0607

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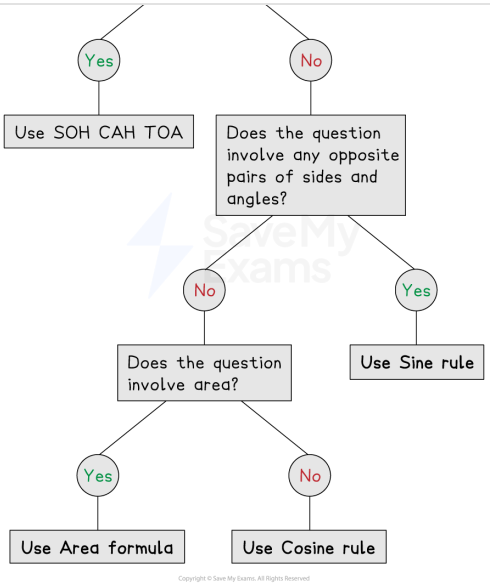
Applications of trigonometry

How do I decide which trig rule to use?

- **Different rules** are required depending on the question
  - You need to be able to decide which is appropriate to use
  - Think about what **information** you **have** and what you want to **find**
- This table summarises the possibilities:

If you know	And you want to know	Use
Two sides and an angle opposite one of the sides	The angle opposite the other side	Sine rule
Two angles and a side opposite one of the angles	The side opposite the other angle	Sine rule
Two sides and the angle **between** them	The third side	Cosine rule
All three sides	Any angle	Cosine rule
Two sides and the angle **between** them	The area of the triangle	Area of a triangle rule

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Flowchart to decide the rule

Can I use multiple trig rules in the same question?

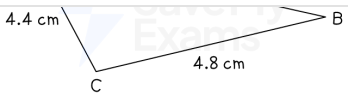
- Harder questions will require you to use **more than one trig rule**
  - For example, you may need the sine rule **followed by** the cosine rule
- The **area formula** only works for an angle between two sides
  - If you are **not** given this setup, you may need to use the sine or cosine rule first
- If it looks like no rule would work, remember that all **angles in a triangle sum to 180**
  - This often helps to find a missing angle

Examiner Tips and Tricks

Look at the number of marks for a question - if it is a lot, you are likely to need more than one trig rule!

Worked Example

Find the area of the triangle below.



Answer:

The area of a triangle can be found using the formula
$A = \frac{1}{2}ab \sin C$

The three side lengths are known , but we need to find an angle in order to calculate the area
Because we know all three sides, any of the angles could be found

Find angle *ABC* using the cosine rule

Cosine Rule: $a^2 = b^2 + c^2 - 2bc \cos A$,
where A is the angle opposite side a

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC) \cos ABC$$
$$4.4^2 = 7.4^2 + 4.8^2 - 2(7.4)(4.8) \cos ABC$$

Rearrange to make $\cos ABC$ the subject

$$4.4^2 - (7.4^2 + 4.8^2) = -2(7.4)(4.8) \cos ABC$$
$$\cos ABC = \frac{4.4^2 - 7.4^2 - 4.8^2}{-2(7.4)(4.8)}$$

Use your calculator to find the value of $\cos ABC$

$$\cos ABC = \frac{487}{592}$$

Use the $\cos^{-1}$ button on your calculator to find the value of ABC

$$ABC = \cos^{-1}\left(\frac{487}{592}\right)$$
$$= 34.65054...$$

Now we can find the area of the triangle using the formula and angle *ABC* as the known angle

$$\text{Area} = \frac{1}{2} \times 4.8 \times 7.4 \times \sin 34.65054...$$
$$= 10.09779...$$

$$\text{Area} = 10.1 \text{ cm}^2 \textbf{ (3.s.f)}$$

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